

20.

Ans: D

$$eV_- + \frac{1}{2}mv^2 = eV_+ + \frac{1}{2}mv_f^2$$

$$e(V_- - V_+) + \frac{1}{2}mv^2 = \frac{1}{2}mv_f^2$$

$$\frac{2e(V_- - V_+)}{m} + v^2 = v_f^2, (V_- - V_+) = -Ex$$

$$v_f = \sqrt{\frac{2e(-Ex)}{m} + v^2}$$

OR

$$\frac{1}{2}mv^2 + W_{\text{electric.force}} = \frac{1}{2}mv_f^2$$

$$\frac{1}{2}mv^2 + (F_{\text{electric.force}} \cdot x) \cos 180^\circ = \frac{1}{2}mv_f^2, [F_{\text{electric.force}} \text{ and } x \text{ opposite direction}]$$

$$\frac{1}{2}mv^2 + -(eE \cdot x) = \frac{1}{2}mv_f^2$$

$$v_f = \sqrt{\frac{2e(-Ex)}{m} + v^2}$$

21.

Ans: C